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# Transitioning into transition metals

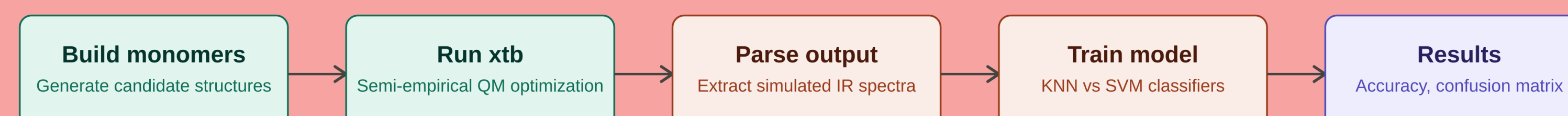
**Introduction:** Metals play essential roles in a huge number of biological processes, and knowing which metal is bound at a given site matters for understanding how these proteins function. Identifying the specific metal bound to a protein or amino acid is a real problem in structural biology and biochemistry.

The current ways of identifying bound metals, such as X-ray or mass spectrometry, tend to be slow, expensive, or require specialized equipment that can be destructive to the metal trying to be extracted. Spectroscopy offers a potentially faster and cheaper alternative, but it raises the question of whether a spectrum alone actually carries enough information to tell metals apart in the first place. Why IR specifically? Vibrational IR spectra are sensitive to the local bonding environment, so a metal bound to an amino acid should be able to be detected by the frequency and eventually predicted. Our research question is can vibrational spectra alone be used to identify the metal systems either, Co, Ni, Cu, or Zn, and whether the vibrational spectral data alone presents enough information to do this? Then our problem becomes a classification problem where machine learning can be applied. I helped build a pipeline that generates simulated IR spectra using xTB for amino acid-metal complexes, then compares two machine learning classifiers, KNN and SVM with an RBF kernel, to see which one can distinguish the four metals more reliably, and how much preprocessing choices like spectral broadening affect that result. Currently, this covers monomer complexes, single amino acid plus metal, with dimer and trimer data available to add later.

**Data:** This dataset consists of computationally generated monomers with each amino acid bound to a transition metal. For each metal amino complex, we run a quantum chemistry calculation called xTB to obtain the vibrational spectrum. This raw xTB output then is parsed to produce simulated IR spectra, which is then broadened using the Gaussian method. To produce a range on intensities across a range of wavenumbers. Each spectrum is treated as a single vector with intensity at each wavenumber being a feature. In total the data contains around 800 labeled spectra distributed across the four metal classes. The IR spectra is then used almost like a fingerprint with the vibrational spectrum used to distinguish the metal bound to the amino acid.

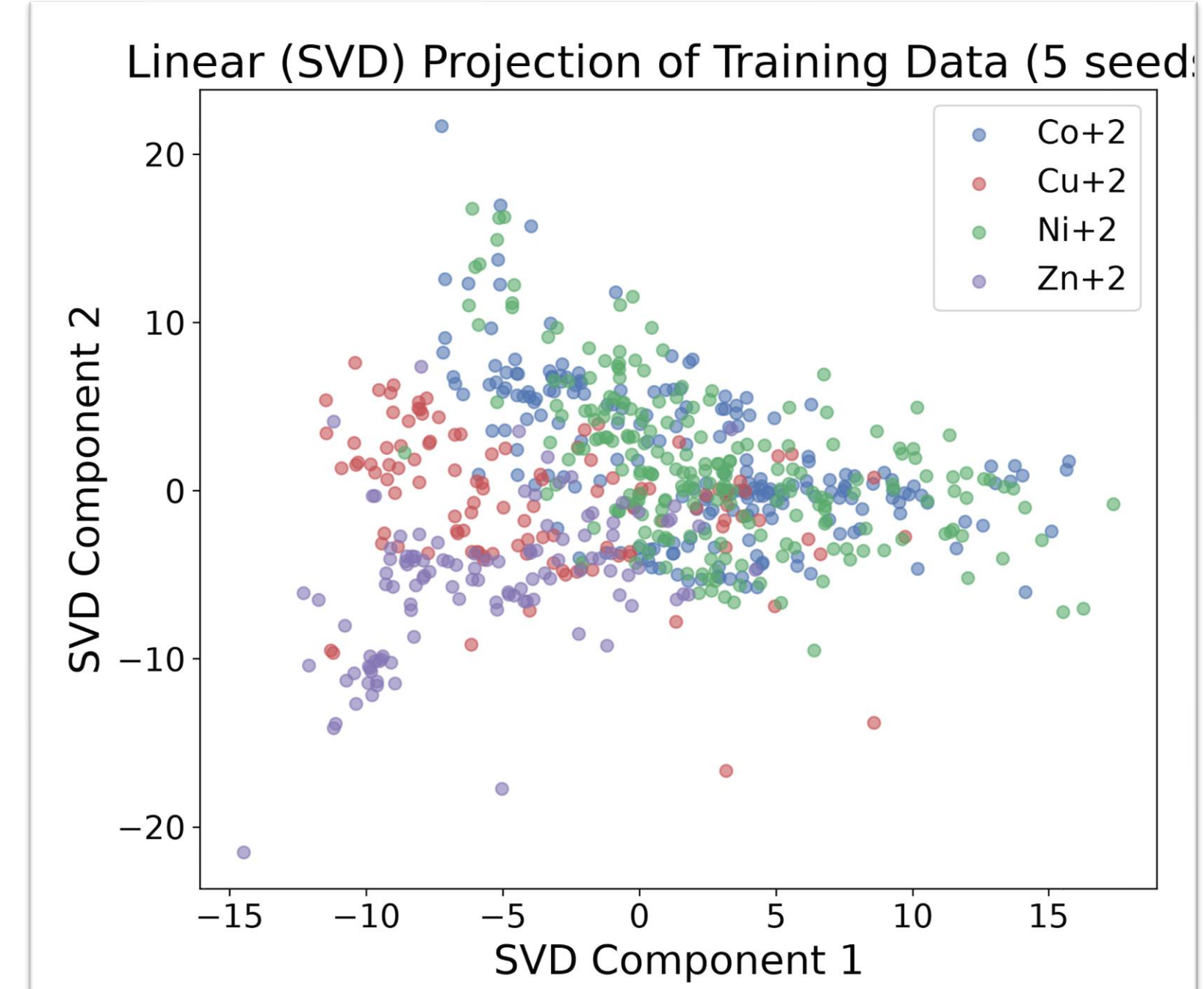
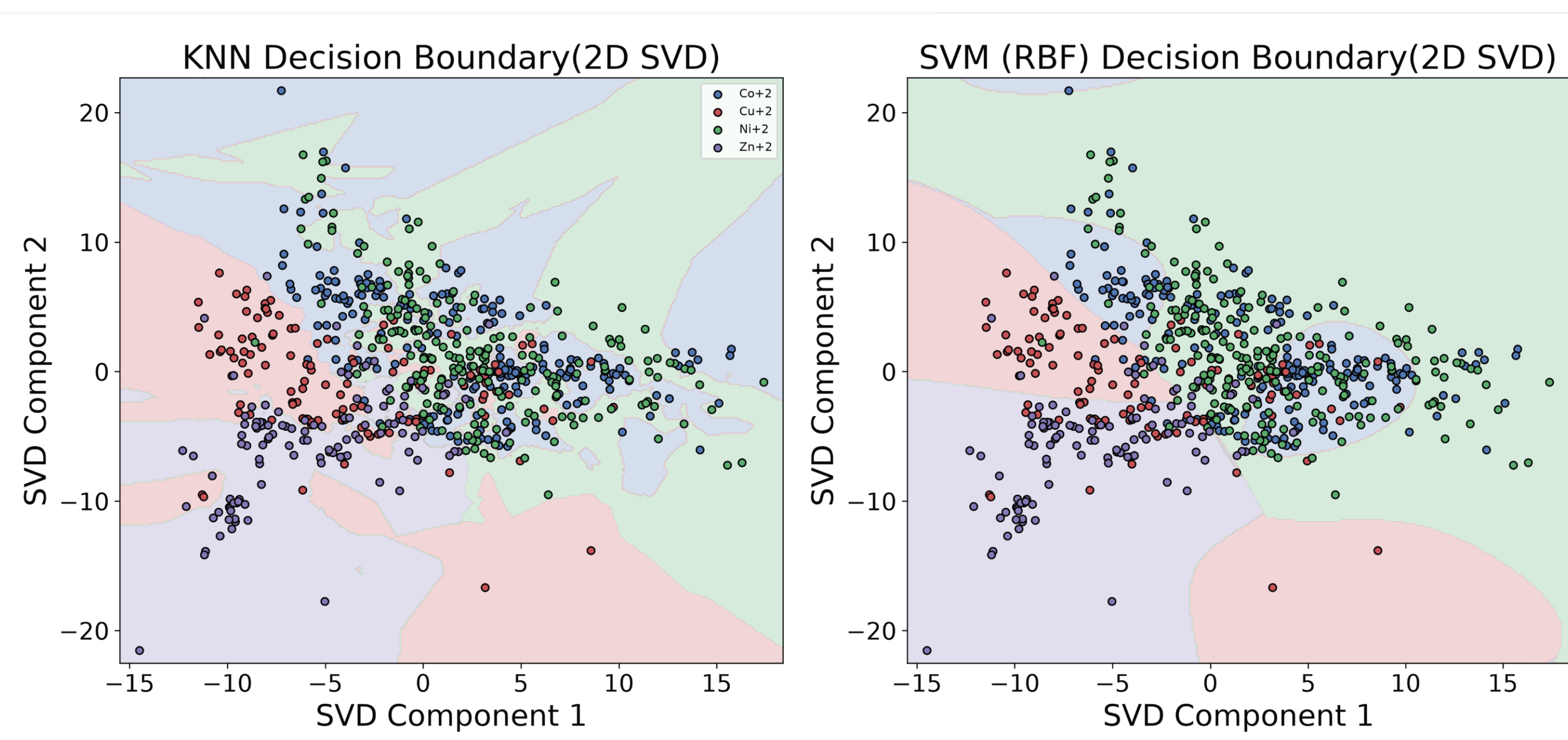
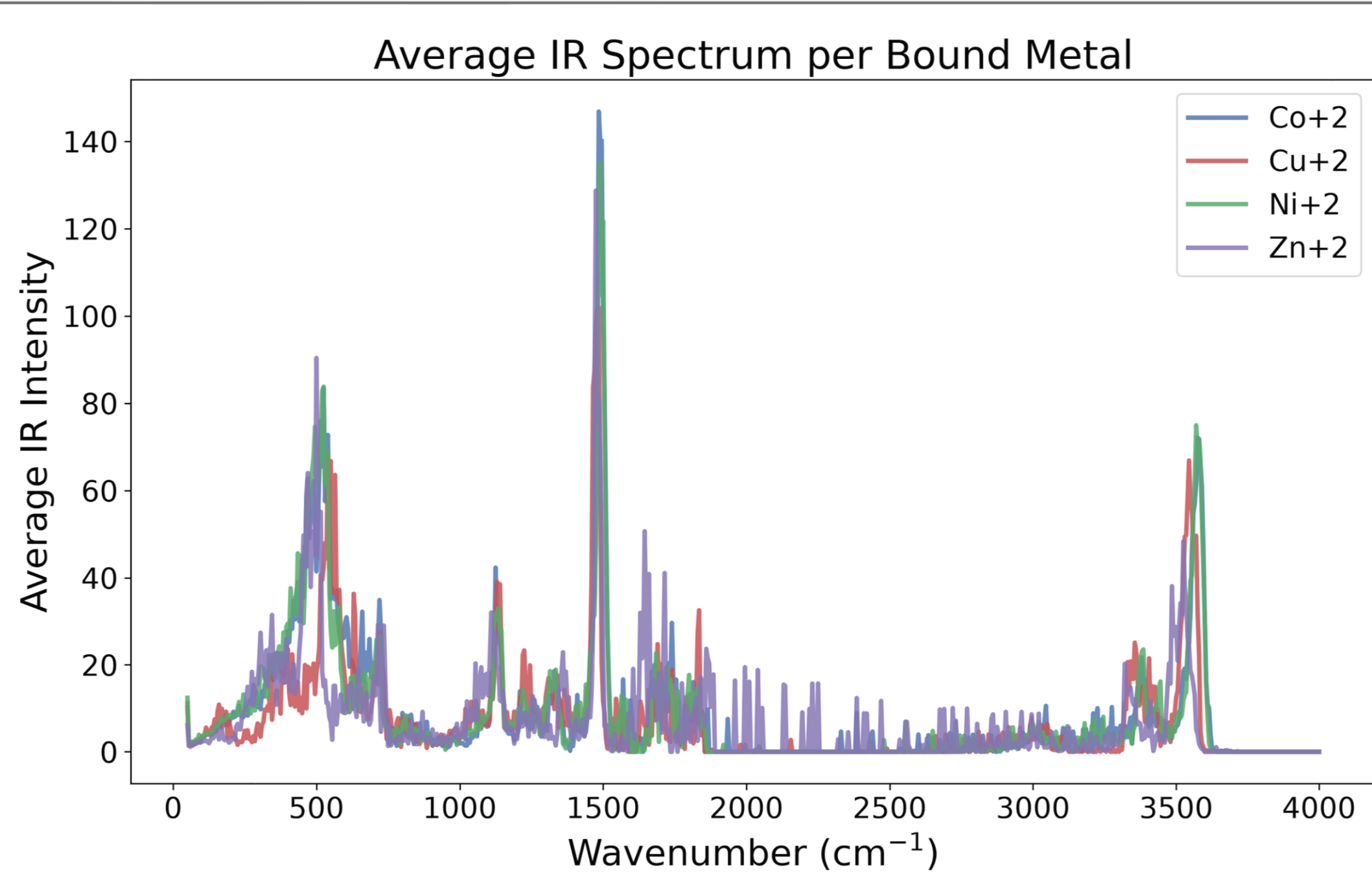
**Pipeline:** 5-step computational process from selecting data generation to classifier performance evaluation.

Step 1: Generate monomer metal-amino acid complexes with our four transition metals (Co, Cu, Ni, Zn)  
Step 2: Each metal complex is passed through xTB a computational chemistry model software. From this we compute the vibrational spectrum for the complex.  
Step 3: The raw output is parsed into simulated IR spectra to produce a list of IR intensities across wavenumbers.  
Step 4: We begin to train our models. We separate our data by grouping metals and conformers to get a good train/test validate split. I decided to compare two classifiers k-nearest neighbor KNN and SVM with RBF kernel.  
Step 5: We get our results from model performance; we have different visualizations showing how well IR spectra alone can be used to predict bound metals.



**Analysis:** To be able to evaluate if IR spectra alone is a good identifier for bound metal complexes two different classifiers were tested and compared. K nearest neighbor and SVM RBF kernel were both trained on the spectral features. We use an SVD scatter plot to better visualize all the spectral features in the data. The 2d visual allows us to see the decision boundaries of each classifier along the training data, this shows how each model partitions the space they have.

Since test train split data can be misleading depending what section of the data is used, I repeated the same test train split across five seeds to get a better average of the results. For each seed, a confusion matrix is made that gets consolidated into an average confusion matrix of each model.



**KNN:** Across 5 seeds, the KNN classifier reached a mean validation accuracy of 57.1% (ranging from 48.2% up to 64.1% depending on the split), correctly classifying spectra across all seeds. Performance varied noticeably by metal: KNN was strongest on Cu<sup>2+</sup> and Co<sup>2+</sup>, but weaker on Ni<sup>2+</sup> and Zn<sup>2+</sup>. The dominant source of error was confusion between Co<sup>2+</sup> and Ni<sup>2+</sup>, Co<sup>2+</sup> spectra were predicted as Ni<sup>2+</sup>, and Ni<sup>2+</sup> spectra were predicted as Co<sup>2+</sup>, together accounting for the majority of KNN's total misclassifications. Because KNN classifies a new spectrum by majority vote among its nearest neighbors in the full spectral feature space, this pattern suggests that Co<sup>2+</sup> and Ni<sup>2+</sup> complexes frequently sit close to one another. In that space their vibrational spectra are locally similar enough that neighboring points often belong to the "wrong" class.

Confusion Matrices (summed across 5 seeds)

|            | KNN  |      |      |      |  | SVM (RBF kernel) (best) |      |      |      |  |
|------------|------|------|------|------|--|-------------------------|------|------|------|--|
|            | Co+2 | Cu+2 | Ni+2 | Zn+2 |  | Co+2                    | Cu+2 | Ni+2 | Zn+2 |  |
| True Metal |      |      |      |      |  |                         |      |      |      |  |
| Zn+2       | 220  | 19   | 103  | 24   |  | 191                     | 0    | 107  | 68   |  |
| Ni+2       | 25   | 124  | 33   | 5    |  | 8                       | 134  | 23   | 22   |  |
| Cu+2       | 133  | 28   | 182  | 16   |  | 94                      | 5    | 206  | 54   |  |
| Co+2       | 35   | 16   | 37   | 106  |  | 1                       | 1    | 17   | 175  |  |

**SVM:** The SVM classifier outperformed KNN, reaching a mean validation accuracy of 63.8% (ranging from 50.9% to 70.5%), it was better at correctly classifying validation spectra summed across all seeds. The improvement over KNN was not uniform across classes. SVM's biggest gain was on Zn<sup>2+</sup>, jumping in accuracy, and it also improved on Cu<sup>2+</sup> and Ni<sup>2+</sup>. Notably, SVM was slightly worse than KNN at identifying Co<sup>2+</sup>, with more Co<sup>2+</sup> spectra misclassified as Zn<sup>2+</sup> than KNN produced. Despite this, the same core weakness persisted: Co<sup>2+</sup> and Ni<sup>2+</sup> remained the most confused pair, with Co<sup>2+</sup> spectra predicted as Ni<sup>2+</sup> and Ni<sup>2+</sup> spectra predicted as Co<sup>2+</sup>. Because this Co<sup>2+</sup> and Ni<sup>2+</sup> confusion keeps showing up under both a locally driven model (KNN) and a kernel-based model (SVM), it points to a genuine overlap in the underlying vibrational signatures of these two metals in this monomer, rather than being a result of either algorithm.

**Transitioning to the future:** One clear next step is bringing a third model into the comparison, most likely a Ridge Classifier or a Partial Least Squares (PLS) model. Both are worth testing for different reasons: Ridge Classifier is a simple linear baseline that would help clarify whether the non-linear boundaries KNN and SVM are drawing are necessary, or whether a much simpler, faster linear model can capture most of the same signal. PLS is also a good fit for this problem since it's a technique built for data of many similar, high-dimensional features like our wavenumber intensities and is widely used in real spectroscopy work for that reason. Adding either model would give a more complete picture of whether KNN and SVM's accuracy reflects genuine non-linear structure in the spectra, or whether a simpler model gets you most of the way there.

A second direction is expanding the dataset beyond single monomers to include dimers and trimers — amino acids bound to a metal in small chains rather than in isolation. This would test whether the vibrational fingerprint approach still holds up as the molecule gets larger and more complex, which matters because real biological and chemical systems rarely involve an isolated single amino acid, usually even more than three. It would also substantially grow the dataset, which could help to see if new dimer/trimer data can be generated with more balanced representation across the four metals or, short of that, resampling or class-weighting techniques could be tested to see whether they close the accuracy gap between the well-represented and under-represented classes.

**Conclusion:** Simulated IR spectra carry real signal for identifying which of four transition metals is bound to a given amino acid, with the best model (SVM, RBF kernel) reaching 63.8% validation accuracy, although Co<sup>2+</sup> and Ni<sup>2+</sup> remain consistently hard to tell apart. This suggests computational vibrational spectroscopy paired with machine learning could serve as a low-cost screening tool for metal amino binding identity, without needing to synthesize or experimentally test every candidate complex. This serves as a potential proof of concept for using IR spectroscopy.

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